LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	19
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need for interlock in AFCS and explanation with schematic
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need for interlock in AFCS and explanation with schematic.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS schematic diagrams; Instructor supervision required for safety regarding electrical systems.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define the term 'interlock' in the context of flight control systems.
(c) Why is system redundancy critical in modern aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic AFCS architecture overview • Definition of interlock mechanism • Prevention of conflicting control inputs • Schematic representation of interlock logic • Operational benefits of interlocking	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of interlock circuit • Signal flow through interlock gates • Failure modes without interlock • Safety implications of interlock failure • Comparison with non-interlocked systems	L & Ds	PPT, PDF	20 min
(c)	• Real-world application of interlock • Maintenance checks for interlock systems • Troubleshooting interlock faults • Integration with other AFCS subsystems • Summary of interlock necessity	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why an interlock is necessary to prevent pilot-induced oscillation.
(b) Describe the function of the interlock gate in the AFCS schematic.
(c) How does the interlock system ensure safe control surface movement?

12. Home Assignment and Follow Up	Prepare a brief note on how the interlock system in the AFCS prepares the trainee for the study of system protection mechanisms in LP 20.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	System protection of AFCS, explanation of comparator, rate trigger system, duplex system, cross coupled feedback system, triplex system, quadruplex and model following system.	Refer to the master syllabus reference list	LP 20

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min